

Electricity Outage Intelligent System in Cameroon

Minimum Viable Product (MVP) & Actors for Analysis

1. MVP Objective

The MVP allows citizens to report a geolocated electricity outage, enables the system to group nearby reports into potential common incidents, tracks incident status, notifies affected users, and provides basic administrative analytics.

2. Citizen Mobile Application

Authentication: Sign up, login, logout, and profile management.

Report an outage: Select complete outage, unstable electricity, or low voltage; capture GPS location automatically; record date/time; optionally add a description or photo.

View nearby outages: Display active outages on an interactive map with status and number of reports.

Confirm an outage: Allow users who are also affected to confirm an existing incident.

Restoration confirmation: Allow users to indicate that electricity has been restored.

3. Intelligent Incident Detection

The system analyzes geographic distance, time between reports, and report volume to determine whether multiple reports may correspond to the same outage event. For the Level 2 MVP, a rule-based geographic and temporal clustering algorithm is sufficient; machine learning is not required.

4. Admin Dashboard

- Dashboard: active outages, total reports, restored outages, and most affected areas.
- Incident management: view reports, locations, details, status, and severity.
- Map: visualize active, recent, and restored incidents.
- Analytics: outages by area, day/week/month, average duration, and report volume.

5. Notifications

- Outage reported in the user's area.
- Multiple reports detected nearby.
- Electricity restoration update.

6. MVP Database Entities

Entity	Key Attributes / Purpose
User	id, name, phone/email, password, role
OutageReport	id, user_id, latitude, longitude, type, description, reported_at, status
OutageEvent	id, area, latitude, longitude, severity, start_time, end_time, status
Notification	id, user_id, title, message, read, created_at
Area	id, name, city, coordinates

7. Actors for System Analysis

Actor	Role	Main Use Cases
Citizen	Primary end user	Register, login, report outage, view outages, confirm outage, report restoration, receive no
System Administrator	Supervises the platform	Login, view dashboard, manage reports/incidents, change status/severity, manage users,

Actor	Role	Main Use Cases
Electricity Service Operator	External operational actor; optional/simulated in MVP	View incidents, affected areas, incident details, update intervention status
Location/Map Service	External technical service	Provide GPS coordinates, geocoding, maps, geographic data

8. MVP Scope — What NOT to Build Yet

- No payment system.
- No advanced predictive AI.
- No direct integration with the electricity provider.
- No IoT or smart-meter integration.
- No automatic image recognition.
- No separate citizen web application if the mobile app already covers the citizen workflow.

9. Final MVP Structure

Citizen Mobile App → **REST API** → **Business Logic** → **PostgreSQL/PostGIS** → **Notification & Map Services** → **Admin Dashboard**

Core workflow: Report → Geolocate → Group → Monitor → Notify → Restore → Analyze.